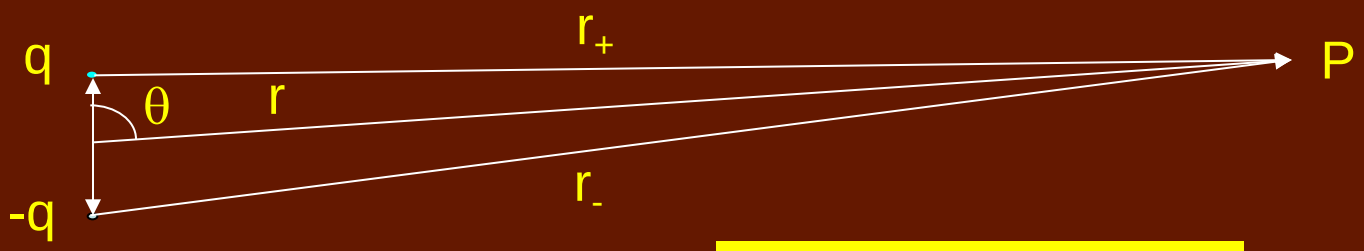
Potential due to a collection of charges



The potential at P due to q and –q is $$V(P) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{r_+} - \frac{q}{r_-}\right]$$

$$r_\pm^2 = r^2 + \left(\frac{s}{2}\right)^2 \mp rs\cos\theta \quad = r^2\left[1 \mp \frac{s}{r}\cos\theta + \frac{s^2}{4r^2}\right]$$

s - separation

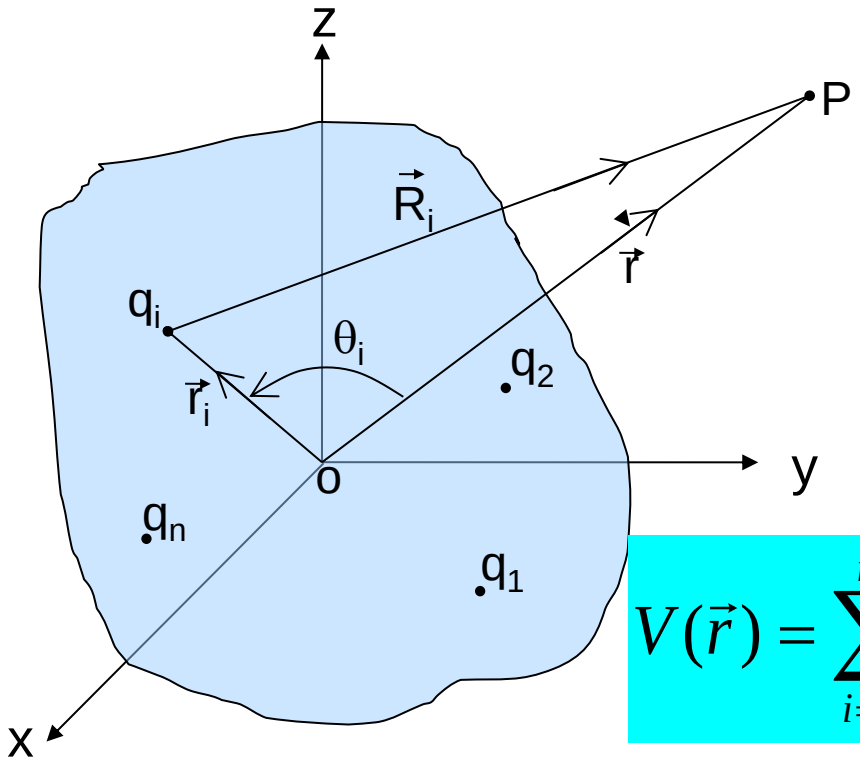
$$\frac{1}{r_\pm} \cong \frac{1}{r}\left[1 \mp \frac{s}{r}\cos\theta\right]^{-1/2} \cong \frac{1}{r}\left[1 \pm \frac{s}{2r}\cos\theta\right]$$

When r >> s,

Then, $$\frac{1}{r_+} - \frac{1}{r_-} = \frac{s}{r^2}\cos\theta \qquad V(P) = \frac{1}{4\pi\varepsilon_0}\frac{qs\cos\theta}{r^2}$$

Thus the potential due to dipole falls as $1/r^2$ instead of $1/r$ for monopole

Multipole expansion of the scalar potential



$$V(\vec{r}) = \sum_{i=1}^{n} \frac{q_i}{4\pi\varepsilon_0 R_i} \quad where\ R_i = \left|\vec{r} - \vec{r}_i\right|$$

In terms of the angle θ_i between the directions of $\vec{r}$ and $\vec{r}_i$ we can write

$$R_i = (r^2 + r_i^{\,2} - 2rr_i\cos\theta_i)^{1/2}$$

$$V(\vec{r}) = \sum_{i=1}^{n} \frac{q_i}{4\pi\varepsilon_0 (r^2 + r_i^{\,2} - 2rr_i\cos\theta_i)^{1/2}}$$

$$R_i = (r^2 + r_i^2 - 2rr_i\cos\theta_i)^{1/2}$$

$$= \left[r^2 \left(1 + \frac{r_i^2}{r^2} - 2\frac{r_i}{r}\cos\theta_i \right) \right]^{1/2}$$

$$= r(1+t)^{1/2} \qquad where \quad t = -2\left(\frac{r_i}{r}\right)\cos\theta_i + \left(\frac{r_i}{r}\right)^2$$

Therefore $$\frac{1}{R_i} = \frac{1}{r(1+t)^{1/2}}$$

Expand $(1+t)^{-1/2}$ as a power series.

$$(1+t)^{-1/2} = 1 - \frac{1}{2}t + \frac{3}{8}t^2 - \frac{5}{16}t^3 + \cdots$$

Neglect all terms with powers more than 2, we get

$$\frac{1}{(1+t)^{1/2}} \approx 1 - \frac{1}{2}\left[-2\left(\frac{r_i}{r}\right)\cos\theta_i + \left(\frac{r_i}{r}\right)^2 \right] + \frac{3}{8}\left[-2\left(\frac{r_i}{r}\right)\cos\theta_i + \left(\frac{r_i}{r}\right)^2 \right]^2$$

$$\approx 1 + \left(\frac{r_i}{r}\right)\cos\theta_i + \frac{1}{2}\left(\frac{r_i}{r}\right)^2 \left(3\cos^2\theta_i - 1\right)$$

$$\therefore V(\vec{r}) = \frac{1}{4\pi\varepsilon_0 r}\sum_{i=1}^{n} q_i + \frac{1}{4\pi\varepsilon_0 r^2}\sum_{i=1}^{n} q_i r_i \cos\theta_i + \frac{1}{4\pi\varepsilon_0 r^3}\sum_{i=1}^{n} \frac{q_i r_i^2}{2}\left(3\cos^2\theta_i - 1\right) + \cdots$$

monopole dipole quadrupole

This is called the *multipole expansion of the potential*.

The monopole, dipole and quadrupole terms fall off as $1/r$, $1/r^2$, and $1/r^3$ respectively.

As we move farther away from the charge distribution the higher order terms become less and less significant.

We can write the potential as

$$V(\vec{r}) = V_m(\vec{r}) + V_d(\vec{r}) + V_q(\vec{r}) + \cdots$$

Alternatively we can write the potential using *Legendre Polynomials* $P_l(x)$. $P_l(x)$ are defined by the expression

$$\frac{1}{\left(1 - 2xy - y^2\right)^{1/2}} = \sum_{l=0}^{\infty} P_l(x) y^l \qquad \left(|x| \le 1, y < 1\right)$$

$P_l(x)$ are the coefficients of y^l in the sum. The first few terms are

$$P_0(x)=1 \quad P_1(x)=x \quad P_2(x)=\frac{1}{2}(3x^2-1) \quad P_3(x)=\frac{1}{2}(5x^3-3x)$$

The other terms can be found by using the *recursion relation*

$$(l+1)P_{l+1}(x)=(2l+1)xP_l(x)-lP_{l-1}(x)$$

$$P_l(1)=1$$

If we take $y = r_i / r$ and $x = \cos\theta_i$ then we can write

$$\frac{1}{R_i}=\frac{1}{r}\sum_{l=0}^{\infty}P_l(\cos\theta_i)\left(\frac{r_i}{r}\right)^l$$

The potential can be written as

$$V(\vec{r})=\frac{1}{4\pi\varepsilon_0}\sum_{l=0}^{\infty}\frac{1}{r^{l+1}}\left[\sum_{i=1}^{n}q_i r_i^{\,l}P_l(\cos\theta_i)\right]$$

We can eliminate θ from the above expression using the relation

$$\cos\theta_i = \frac{\vec{r}\cdot\vec{r}_i}{rr_i} = \hat{r}\cdot\left(\frac{\vec{r}_i}{r_i}\right) = \frac{l_x x_i + l_y y_i + l_z z_i}{r_i}$$

Where l_x, l_y, and l_z are the direction cosines of the position vector $\vec{r}$, of P and x_i, y_i, and z_i are the rectangular coordinates of the location of the charge q_i.

Let us now consider the terms one by one for convenience.

Monopole term

$$\sum_{i=1}^{n} q_i = Q$$

$$V_m(\vec{r}) = \frac{Q}{4\pi\varepsilon_0 r}$$

This is the dominant term in the potential. At far away distance the whole charge distribution will behave as if it is were a point charge.

Q is called the *monopole moment* of the charge distribution.

For continuous charge distribution $Q = \int_{V'} \rho(\vec{r}')d\tau'$

Dipole term

$$\sum_{i=1}^{n} q_i r_i \cos\theta_i = \sum_{i=1}^{n} q_i (l_x x_i + l_y y_i + l_z z_i)$$

$$= l_x \left(\sum_i q_i x_i \right) + l_y \left(\sum_i q_i y_i \right) + l_z \left(\sum_i q_i z_i \right)$$

$$= \hat{r} \cdot \left(\sum_{i=1}^{n} q_i \vec{r}_i \right)$$

The term inside the bracket does not involve the location of the field point and it is the property of the charge distribution only. This term is called the *dipole moment* $\vec{p}$.

$$\vec{p} = \sum_{i=1}^{n} q_i \vec{r}_i$$

$$\sum_{i=1}^{n} q_i r_i \cos\theta_i = \hat{r} \cdot \vec{p} = l_x p_x + l_y p_y + l_z p_z$$

$$V_d(\vec{r}) = \frac{\vec{p} \cdot \hat{r}}{4\pi\varepsilon_0 r^2} = \frac{\vec{p} \cdot \vec{r}}{4\pi\varepsilon_0 r^3}$$

If the point is very far away and if the monopole moment vanishes, then dipole moment will be the dominant term in the expansion of potential.

The Quadrupole term

This term in complicated

$$r_i^2\left(3\cos^2\theta_i - 1\right) = 3\left(\hat{r}\cdot r_i\right)^2 - r_i^2$$

$$= 3(l_x x_i + l_y y_i + l_z z_i)^2 - r_i^2(l_x^2 + l_y^2 + l_z^2)$$

Grouping the factors together

$$= l_x^2(3x_i^2 - r_i^2) + l_y^2(3y_i^2 - r_i^2) + l_z^2(3z_i^2 - r_i^2) +$$

$$6l_x l_y x_i y_i + 6l_y l_z y_i z_i + 6l_z l_x z_i x_i$$

Put this big term back in the main equation for potential

Rearrange the terms similar to this in a specific format

$$6l_x l_y x_i y_i = 3l_x l_y x_i y_i + 3l_y l_x y_i x_i$$

You end up with a symmetric form of equation

$$\sum_i q_i r_i^2 \left(3\cos^2\theta_i - 1\right) =$$

$$l_x^2 \sum_i q_i (3x_i^2 - r_i^2) + l_x l_y \sum_i q_i 3x_i y_i + l_x l_z \sum_i q_i 3x_i z_i +$$

$$l_y^2 \sum_i q_i (3y_i^2 - r_i^2) + l_y l_x \sum_i q_i 3y_i x_i + l_y l_z \sum_i q_i 3y_i z_i +$$

$$l_z^2 \sum_i q_i (3z_i^2 - r_i^2) + l_z l_x \sum_i q_i 3z_i x_i + l_z l_y \sum_i q_i 3z_i y_i$$

Each term in the expression is a product of something that depends only on the field point (its direction) and a quantity that depends only on the details of its charge distribution

We define here a new quantity Q_{jk}, which are called the components of the quadrupole moment tensor

$$Q_{jk} = \sum_{i=1}^{N} q_i \left(3 j_i k_i - r_i^{\,2} \delta_{jk} \right) \qquad (j,k = x,y,z)$$

j and k can independently be x, y or z

δ_{jk} *is the Kronecker delta symbol*

$$\delta_{jk} = \begin{cases} 1 & \text{if } j = k \\ 0 & \text{if } j \neq k \end{cases}$$

There a total of nine Q_{jk} in our main equation in the previous page

$$Q_{xx} = \sum_i q_i \left(3 x_i^{\,2} - r_i^{\,2} \right)$$

$$Q_{xy} = \sum_i q_i \, 3 x_i y_i$$

With this we can write the whole term in a compact form

$$\sum_i q_i r_i^{\,2}\left(3\cos^2\theta_i - 1\right) =$$

$$= l_x^{\,2} Q_{xx} + l_x l_y Q_{xy} + l_x l_z Q_{xz} +$$

$$l_y^{\,2} Q_{yy} + l_y l_x Q_{yx} + l_y l_z Q_{yz} +$$

$$l_z^{\,2} Q_{zz} + l_z l_x Q_{zx} + l_z l_y Q_{zy}$$

$$= \sum_{j=x,y,z} \sum_{k=x,y,z} l_j l_k Q_{jk}$$

Incorporating this into the quadrupole term of the potential, we get

$$V_Q(\vec{r}) = \frac{1}{4\pi\varepsilon_0 r^3} \cdot \frac{1}{2} \sum_{j=x,y,z} \sum_{k=x,y,z} l_j l_k Q_{jk}$$

If the point P is very far away and if both monopole moment Q and dipole moment **p** are zero, the above term will be the leading term in the expansion of Potential and Quadrupole moment tensor Q_{jk} will be the dominant feature of the charge distribution

We can also write the above in terms of coordinates rather than direction cosines

$$l_x = \frac{x}{r} \qquad l_y = \frac{y}{r} \qquad l_z = \frac{z}{r}$$

$$V_Q(\vec{r}) = \frac{1}{4\pi\varepsilon_0 r^5} \cdot \frac{1}{2} \sum_{j=x,y,z} \sum_{k=x,y,z} jk Q_{jk}$$

For a continuous charge distribution

$$Q_{jk} = \int_{V'} \rho(\vec{r}\,') \left(3 j' k' - r'^{2} \delta_{jk}\right) d\tau'$$

Although there were a total of nine Q_{jk} , there are actually fewer independent ones

We know $Q_{xy} = Q_{yx}$

$$Q_{kj} = Q_{jk} \qquad (j \neq k)$$

Thus Quadrupole moment tensor is an example of symmetric Tensor

Sum of the diagonal components, those with j = k

$$Q_{xx} + Q_{yy} + Q_{zz} = \sum_i q_i \left[\left(3x_i^{\,2} - r_i^{\,2}\right) + \left(3y_i^{\,2} - r_i^{\,2}\right) + \left(3z_i^{\,2} - r_i^{\,2}\right)\right] = 0$$

$$Q_{xx} + Q_{yy} + Q_{zz} = 0$$

If the charge distribution has sufficient symmetry, the number of independent components reduce further

Consider the case of axial symmetry (cone, cylinder)..

Choose the z-axis to be along this axis and designate the elements in this case by $Q^{a}{}_{jk}$

For every charge q/ at (x/, y/, z/) there will be a charge of the same value at (-x/, y/, z/)

Contribution of this pair to $Q_{xy}{}^{a}$ will be $3q'x'y' + 3q'(-x')y' = 0$

Since all of the charges can be paired in this manner $Q_{xy}{}^{a} = 0$

Same argument holds for the rest of the off-diagonal elements

$$Q_{jk}{}^{a} = 0$$

So , there are only three components.

$$Q_{xx} + Q_{yy} + Q_{zz} = 0$$

There is no real distinction between x and y $Q_{xx}{}^{a} = Q_{yy}{}^{a}$

Therefore $2Q_{xx}{}^{a} + Q_{zz}{}^{a} = 0$

or $$Q_{xx}{}^{a} = Q_{yy}{}^{a} = -\frac{1}{2} Q_{zz}{}^{a}$$

So, there is only one independent component of the quadrupole moment characteristic of the charge distribution

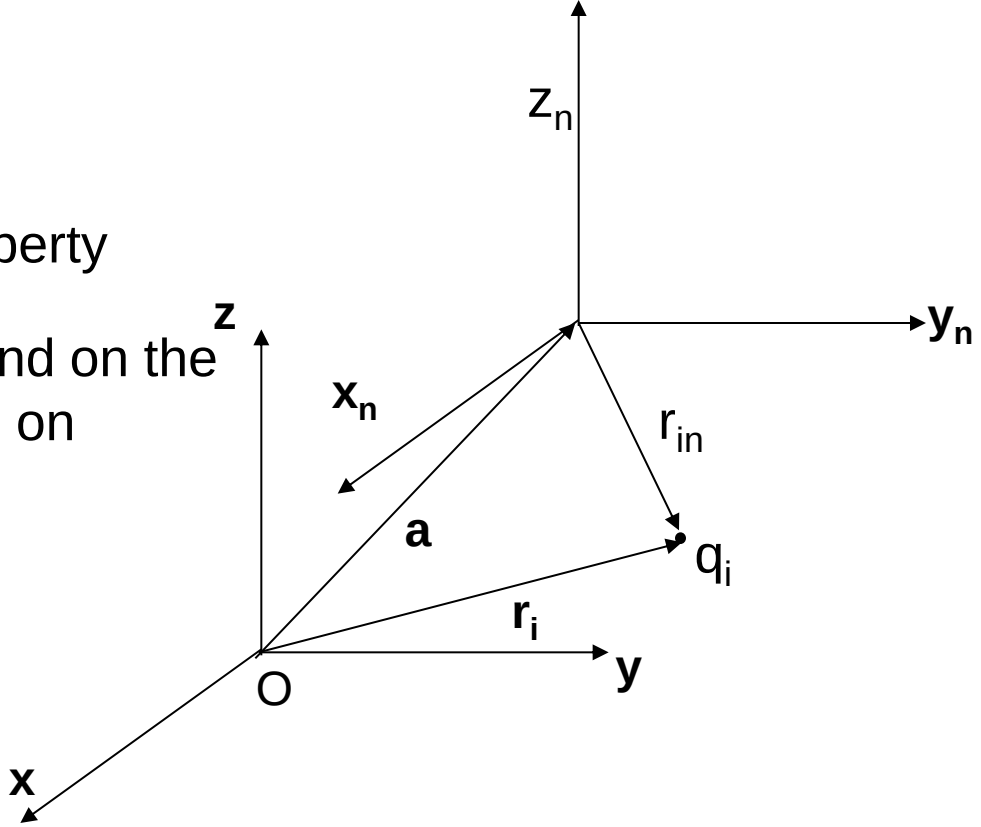
In this case quadrupole term will become

$$V_Q{}^{a}(\vec{r}) = \frac{Q^a}{4\pi\varepsilon_0} \frac{\left(3l_z{}^2 - 1\right)}{4r^3} = \frac{Q^a}{4\pi\varepsilon_0} \frac{\left(3\cos^2\theta - 1\right)}{4r^3}$$

Effects of change in the origin

Monopole moment is a unique property

But dipole and quadrupole terms depend on the choice of the origin, since they depend on The absolute value of the origin



$$\vec{r}_{in} = \vec{r}_i - \vec{a}$$

New value of dipole moment is

$$\vec{p}_n = \sum_i q_i \vec{r}_{in} = \sum_i q_i \vec{r}_i - \vec{a} \sum_i q_i = \vec{p} - Q\vec{a}$$

$$\vec{p}_n = \vec{p}$$ When the monopole moment vanishes

Field of Pure Dipole

$$V_{dip}(r,\theta) = \frac{\hat{r}\cdot\vec{p}}{4\pi\varepsilon_0 r^2} = \frac{p\cos\theta}{4\pi\varepsilon_0 r^2}$$

$$E_r = -\frac{\partial V}{\partial r} = \frac{2p\cos\theta}{4\pi\varepsilon_0 r^3}$$

$$E_\theta = -\frac{1}{r}\frac{\partial V}{\partial \theta} = \frac{p\sin\theta}{4\pi\varepsilon_0 r^3}$$

$$E_\phi = -\frac{1}{r\sin\theta} = \frac{\partial V}{\partial \phi} = 0$$

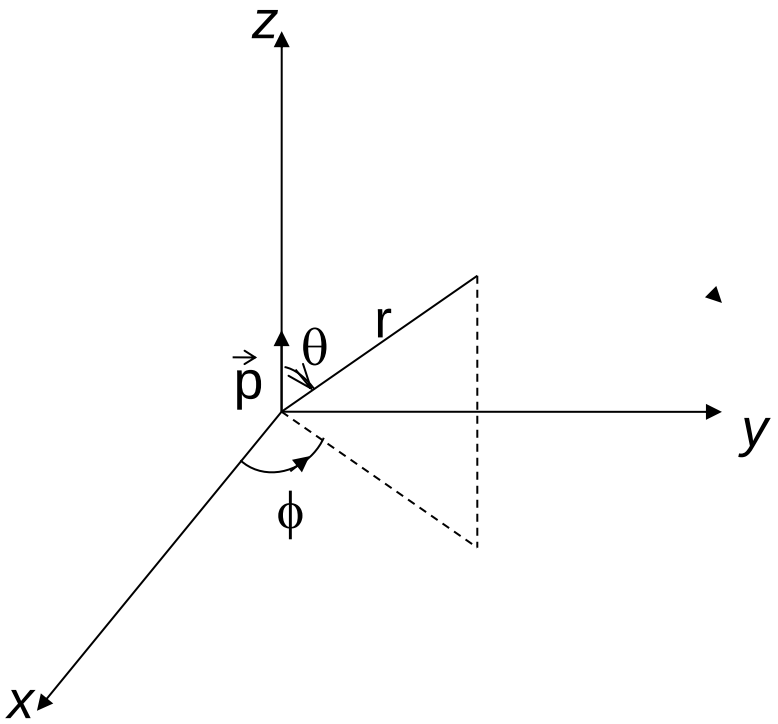
$$E_d(r,\theta) = \frac{p}{4\pi\varepsilon_0 r^3}\left[2\cos\theta\,\hat{e}_r + \sin\theta\,\hat{e}_\theta\right]$$

Coordinate Free form

$$E_{dip}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{1}{r^3}\left[3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}\right]$$

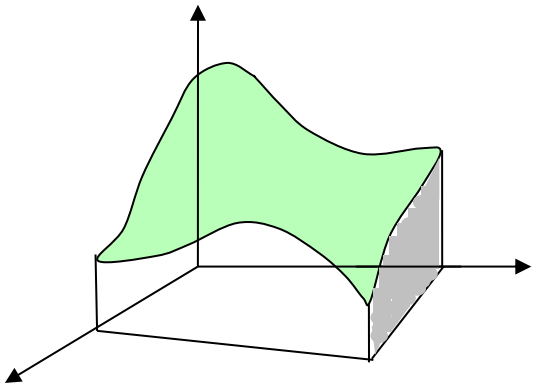
$$3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p} = 3p\cos\theta\,\hat{r} - p\cos\theta\,\hat{r} + p\sin\theta\,\hat{\theta}$$

$$= 2p\cos\theta\,\hat{r} + p\sin\theta\,\hat{\theta}$$



Solutions to Poison's and Lapalce's equations

Refer Lecture Notes 1 for the solutions to Laplace equations in one dimension for cartesian, spherical and cylindrical coordinates



$$\nabla^2 V = -\frac{\rho}{\varepsilon_0}$$ ----------Poisson's equation

In charge free region, $\nabla^2 V = 0$ ----------Laplace Equation